shifted left by 2 cities; in fact, the cost of the two tours is the same (because they involve the same edges).

There are 6 equivalent sequences that can be derived from the one we gave as solution, one for each city used as starting point.

We can therefore fix in advance the city from which the tour starts, knowing that this won't affect the result, but it will cut the number of permutations we need to examine: from 720 we go down to only 120 – not bad, and it will be an even better gain for larger graphs.

Moreover, consider that this will work well with the specific instance of the problem we need to solve for our ecommerce company: we always need to start (and end) our tours at the warehouse, where the goods to deliver are loaded on the trucks.

If we set, for instance, `New Carthage` as the starting point, there might still be several equally good solutions (if multiple sub-paths have the same total cost), but if the graph is undirected there will only be, at most, two equivalent solutions:

```
[New Carthage, Happy Harbor, Gotham City,
Metropolis, Opal City, Civic City]
```

```
[New Carthage, Civic City, Opal City, Metropolis,
Gotham City, Happy Harbor]
```

That's because if edges have the same cost in both directions, we can travel through a simple cycle (a tour) in